2. Requirement Analysis Phase

Functional and non-functional requirements

Functional Requirements

- Q&A; functionality through the /qa endpoint.
- Concept explanation through the /explain endpoint.
- Quiz generation through the /quiz endpoint.
- Text summarization through the /summarize endpoint.
- Learning recommendations through /learn/recommendations.
- Frontend task selection through a dropdown containing Explain, QnA, Quiz, Summary and Recommend Path.
- Form submission from the frontend to the FastAPI backend and display of the result.

Non-Functional Requirements

- Simple and responsive web interface.
- Readable and concise educational output.
- Lightweight infrastructure suitable for local use.
- Secure handling of the Google Gemini API key.
- Error handling for invalid or incomplete AI responses.
- Maintainable modular code with separate files for each major function.

Software Prerequisites

- Python 3.10+.
- FastAPI.
- HTML and CSS.
- Google Gemini API key.
- Uvicorn.
- Jinja2.

System Inputs

User questions, topics, passages and learning requests.

System Outputs

Answers, simplified explanations, three-question MCQ quizzes, summaries and structured learning recommendations.

Requirement Traceability

The requirements above map directly to the modules and endpoints documented in the supplied project document.